

Byeharu — Project Status

Plain-English status of what is built, live, dark, and next · main a947c8d · production migration head 0072 · 2026-07-01

Is your summary correct?

Yes — your 10-section summary is accurate. Every major claim checks out against the repository and production state I verified: the milestones, the main-ship transition, Open-Space Navigation, ports/docking, the first-ship (PORT-ENTRY) backend, the dark coordinate system, and the operational facts (head 0072 , verifier merged but not yet run, coordinate travel dark). This document keeps your structure, explains each part in simple terms, and **adds the few things worth making explicit** (highlighted in green boxes).

0. The 30-second version

Byeharu is a **server-authoritative PvE space-strategy game** — the server decides everything that matters (where your ship is, who wins a fight, what you earn); the browser only shows results and asks the server to act. That makes it cheat-resistant by design.

Today a player can sign up, explore the world map, run the original "base sends a disposable fleet to fight pirates" loop, produce units, and operate a **persistent main ship** that travels **port-to-port**, docks at real ports, and shows its status. The newest capability — **commissioning a brand-new player's first ship and "finishing docking" for legacy ships** — exists and is **deployed on the server**, but does **not yet have a player-facing button**. Free "fly to any X/Y coordinate" travel is fully built but deliberately **switched off (dark)** until a separate go-live decision. **Trading is fully designed but not built.**

Status words used below: LIVE players can use it now · BUILT exists on server/code (maybe behind a gate) · DARK built but switched off on purpose · NOT BUILT still to do.

1. Foundation milestones (M1–M7) — what was built first

Milestone	What it is, in plain terms	Status
M1 — Scaffold & identity	The app shell (React/Vite/TypeScript), a state store, the Supabase backend, and full sign-up / sign-in / session + profiles with protected routes. <i>You can make an account and stay logged in.</i>	BUILT
M2 — World map	The universe is organized sector → zone → location , each with coordinates, type, and difficulty, served by a read-only map function. <i>You can look at the world.</i>	BUILT
M3 — Bases, fleets, movement	Home bases, resources, unit types, the player-initialization that gives a new account its starting base/units, and the send/return movement engine. <i>The original "send a fleet out and bring it back" loop.</i>	BUILT
M4 — Combat loop	Pirate encounters on arrival, tick-based combat with waves, retreat timing, the return trip, and rewards granted only when the fleet gets home, plus combat reports. <i>Real PvE missions with risk and payoff.</i>	BUILT
M4.5 — UX rules	A serial production queue, cancel/refund behavior, clearer travel wording, a coordinates label, and a completed-actions history. <i>The loop becomes understandable.</i>	BUILT
M5 — Living world	Locations and zones have changing danger/pressure that drifts and decays back toward a baseline. <i>The map feels alive, not static.</i>	BUILT
M6 — Frontend clarity	Danger display + preview-before-you-send, fleet lifecycle UI, combat-round logs, a reports page, and time/duration UI. <i>You can read what's happening.</i>	BUILT
M7 — Production system	Unit costs, build orders, a train-units command, the queue processor, and a Train Ships screen. <i>You can produce the old fleet units.</i>	BUILT

2. Main-ship transition — moving away from disposable fleets

The long-term game is about a **persistent main ship** (a lasting asset with identity, HP, captain/module direction) rather than throwaway metal fleets. The transition was done **additively** — the old fleet loop keeps working at every step.

Phase	Plain meaning	Status
Phase 10 — Foundation	The persistent main-ship model and the captain/ship direction. <i>Your progression object is a ship, not disposable metal.</i>	BUILT
10C — Send path	The main ship can be sent to a named world location, independent of the old fleet loop.	BUILT
10D — UI overlay	Main-ship controls in the UI (behind feature gates).	BUILT
10E — Isolation	Main ships are hidden/de-emphasized in the legacy Fleets view so the two systems don't confuse players.	BUILT

10F — Repair / safelock	A destroyed main ship can be repaired/recovered, so a player can never permanently soft-lock after a loss. <i>(Today this repair is free/instant — a safety net, not final gameplay.)</i>	BUILT
10H — Command Center	See the main ship's status, HP, destination, countdowns, and state.	BUILT
Present-to-move	The ship can move location → location without first returning home, enabling real travel chains.	BUILT

Worth adding (not in your table)

- **"One main ship per player" is a deliberate *starter* rule, not the final design.** The database enforces one ship per account today (a uniqueness rule on `player_id`). The intended future is **several** persistent ships per player, freely assigned to any activity — the contracts were shaped now so that future needs no rewrite.
- **High-stakes model is designed, not built.** The long-term plan is that ships *can be permanently destroyed* in combat, with an **emergency weak starter ship** granted only when a player hits *zero* usable ships (so the account never locks). That ships **with** destruction, which depends on main-ship combat (still future). Today's free repair (10F) is the interim safety net.

3. Open-Space Navigation (OSN) — the strategic travel system

OSN gives the ship a single, authoritative **physical state in the universe** and makes movement **server-controlled, auditable, and safe under concurrency**.

Piece	Plain meaning	Status
Spatial states + coordinates	The ship is always exactly one of: <code>home</code> , <code>at_location</code> (docked), <code>in_transit</code> , <code>in_space</code> , or <code>destroyed</code> — with real coordinates. No contradictory states allowed.	BUILT
Transition rules + resolver	One audited place decides "where is the ship and can it do X," preventing unsafe transitions.	BUILT
Movement schema + core	Movement records (origin/target, times, speed, reason, idempotent command receipts) with locking, validation, bounds, zero-distance and travel-time-cap checks. <i>Strategic moves are auditable and can't be duplicated by a double-click.</i>	BUILT
Arrival processor	A scheduled job settles completed moves on the server — the browser never decides arrival.	BUILT
Destruction hardening	Only a trusted server path can write the "destroyed" state.	BUILT
Raw coordinate command	"Move to X,Y" exists on the server — the technical base for free-space travel.	DARK
Coordinate frontend foundation	A stable fixed-coordinate transform + rendering groundwork (so a coordinate map won't have unstable math).	DARK
Stop mid-travel	"Stop the ship safely while flying" exists on the server.	DARK

4. Ports, docking, and the current travel loop

Capability	Plain meaning	Status
Canonical space anchors	Every port/location has one authoritative coordinate (its "anchor") — the single source of truth for where it is.	BUILT
Dock-at-location	The ship can arrive at an eligible port and become <code>at_location</code> (docked) — OSN travel connects to real port presence.	BUILT
Docking integrity	Docking re-checks the target is legal, the anchor still matches, and the fleet/presence/movement state is coherent — no "docked at an impossible place."	BUILT
Port-to-port travel	Travel from one eligible port to another (<code>command_main_ship_space_move_to_location</code>).	LIVE
Starter ports	Haven Reach, Slagworks Anchorage, Driftmarch Waypost — the first public port network.	LIVE
Dock-service read surface	See the services at your current dock (<code>get_my_current_dock_services()</code> + Dock Services UI). Today only "Docking" is offered.	LIVE
Legacy arrival compatibility	Old named-location arrivals can end as <code>legacy_present</code> , so the old system coexists with OSN.	BUILT
Same-location normalization	<code>normalize_main_ship_dock()</code> upgrades a valid <code>legacy_present</code> ship to canonical docked state without moving it .	BUILT (server) · UI not built

5. First-ship / Port-Entry work (the newest milestone)

This is the bridge that makes a brand-new player able to *have* a docked ship — the prerequisite for Trading. It was deployed to production as migration `0072`.

Item	Plain meaning	Status
First-ship commission rule	A player with no ship can claim exactly one first ship.	DEPLOYED
Starter location	The first ship appears already docked at Haven Reach (a spawn placement, not a "home port").	DEPLOYED
Commission command	<code>commission_first_main_ship()</code> atomically creates the ship + fleet + docked presence in one safe step.	DEPLOYED
Duplicate/concurrency safety	Two simultaneous claims produce exactly one ship; the loser is a safe no-op. <i>Proven with a real two-session race test.</i>	DEPLOYED + PROVEN
Normalize command	<code>normalize_main_ship_dock()</code> turns a valid legacy dock state into the canonical docked state.	DEPLOYED
Private writer	<code>port_entry_commission_writer(uuid)</code> — server/service-role-only; players can never call it directly.	DEPLOYED
"Claim first ship" UI	The onboarding button/flow for a new account.	NOT BUILT
"Finish docking" (normalize) UI	A player-facing action for legacy ships.	NOT BUILT

Worth adding

- **The first-ship engine has its own independent production verifier — built and merged, but not yet run.** A strictly read-only, production-gated tool (`port-entry-1-production-verify`) can prove that production actually contains the right function bodies, ACLs, flags, and the **complete authenticated RPC surface**. It is `workflow_dispatch` -only with **zero runs** so far — running it is the immediate, human-approved next step (see §10).

6. Coordinate travel — built, but dark on purpose

Area	State
Empty-space coordinate movement (command, arrival, settlement)	BUILT
Server feature flag <code>mainship_coordinate_travel_enabled</code>	false
Runtime capability <code>coordinate_travel_available</code>	false
Coordinate UI	Hidden
Raw coordinate command while flag is false	Server-rejected
Port-to-port movement	Unaffected / still LIVE

Worth adding — the frontend is already wired

The coordinate UI is not just "hidden" — the **frontend on the live site already reads the server's `coordinate_travel_available` at runtime** and is built fail-closed. The old compile-time on/off switch was retired. **Practical effect:** when the server flag is eventually flipped to `true`, the coordinate targeting UI **lights up automatically with no new frontend deploy**. Until then it stays invisible and the raw command stays rejected. Activation remains a **separate, gated decision** with its own post-enable verification.

7. What players can actually do today

Activity	Reality
Create / log in to an account	Works
View map & world locations	Works
Use the legacy base/fleet system & produce units	Works
Send fleets, fight pirates, retreat, return, get reports/rewards	Works
Operate the persistent main ship; see status/HP/destination/countdown	Works

Repair a destroyed main ship	Works
Travel the main ship port-to-port & dock; view dock services	Works
Fly to arbitrary X/Y coordinates	Not live (dark)
New account claims its first ship through the UI	Not yet (server ready, no button)
Legacy ship "finish docking" through the UI	Not yet (server ready, no button)
Buy/sell goods, cargo, credits, market	Not built (designed)
Mining / exploration / OSN tactical combat	Future

8. Trading V1 — decided, not yet implemented

The economy's first version is **fully designed and calibrated** but no code exists yet.

Component	Decision	Status
Eligibility	Must own a valid main ship docked at an active market-capable port	Designed
Cargo model	Hybrid — account inventory <i>plus</i> a per-ship trade hold	Designed
Wallet	Lazy account-credits wallet	Designed
Market offers	Server-owned offers; server-authoritative price/availability	Designed
Buy/sell	Idempotent commands + receipts	Designed
UI & backend (wallet, cargo, offers, settlement)	Port market screen, cargo, buy/sell	Not built

9. Operational & verification state

Production migration head	0072
Current main	a947c8d49897b1abf9157a7c43a3ef11e1a81e0a
PORT-ENTRY server functions	Deployed (commission + normalize + private writer)
PORT-ENTRY production verifier tooling	Merged to main (PR #62)
Verifier tooling Build check	Passed — run 28453279886
Production verifier workflow	Exists, dispatch-only, zero runs
Production verifier result	Not yet run
Deployment triggered by the verifier-tool merge	No (tooling-only; no migration changed)
Coordinate activation	Not authorized / not performed
Trading implementation	Not started
PORT-ENTRY player UI	Not started

Worth adding — the engineering discipline

Every backend slice in this sequence shipped through the same controlled path: a **disposable real-Postgres proof** on the actual migration chain → an **independent pre-merge review** → a **human-approved** protected merge → a **human-gated production deploy** → (for the catalog) a **read-only production verifier**. Concurrency was proven with a genuine two-session race; production gates and PR approvals are human-controlled. This is why the project can move fast *and* stay safe.

10. What should happen next (recommended order)

#	Work	Needs a human gate?	Why
1	Run the existing read-only PORT-ENTRY production verifier	Yes — approve the production environment gate when it pauses	Independently confirms production has the right RPC bodies, ACLs, flags, and the complete authenticated RPC surface. Closes the loop on the just-deployed first-ship engine.
2	Build the "Claim First Ship" onboarding UI	No special prod approval to build/review	Turns the backend-only first-ship capability into a real, playable new-player flow.

3	Build the "Finish docking" (normalize) UI	No special prod approval to build/review	Lets existing/legacy ships join the modern port system safely.
4	Merge/deploy the PORT-ENTRY UI (normal review + checks)	Yes — normal merge/deploy governance	Turns the first real port-entry loop on for players.
5	Begin Trading V1 backend + UI	Normal governance	The next core economy loop. Does not need coordinate activation first.
6	Coordinate-travel activation	Yes — separate future decision + post-enable verification	Not required for ports or Trading V1; flip the flag only when its own activation/verification is approved.

The honest bottom line

The game is **not stuck on missing world systems**. It already has: accounts · map · bases · fleets · PvE combat · production · persistent main ships · port-to-port travel · docking · dock services · first-ship server capability · safe legacy-to-port normalization · and a fully-built coordinate system that's dark and ready for later. The one missing piece in the immediate path is that the new port-entry capability is still **backend only** — there's no player-visible onboarding yet. So the next active development is exactly as you said: **build the "Claim First Ship" flow and the legacy dock-normalization flow, then start Trading V1** — with the production verifier run as the immediate gated checkpoint first.

Glossary (quick, plain definitions)

Term	Means
Server-authoritative	The server decides outcomes; the browser only displays and requests. Cheat-resistant.
Main ship	Your persistent, lasting ship — the long-term progression object (vs. disposable fleets).
OSN	Open-Space Navigation — the system that gives the ship one true physical state + safe, server-run movement.
at_location / docked	The ship is parked at a real port; the prerequisite for trading there.
legacy_present	An older "present at a place" state that can be upgraded to canonical docked via <i>normalize</i> .
Port-to-port	Travel between named, eligible ports (live). Different from free X/Y coordinate travel (dark).
Commission	Create a brand-new player's first ship, already docked at Haven Reach.
Normalize	Convert a legacy docked ship to the modern canonical docked state — in place, no movement.
Dark / flag	Code exists but is switched off by a server flag, on purpose, until a separate go-live.
Idempotent	Doing the same command twice has the same effect as once — no duplicates from double-clicks or retries.
Migration head	The latest database change applied to production (currently 0072).
Production gate	A required human approval before anything touches the live database.